

ORIGINAL ARTICLE

An examination of the effectiveness of video modelling intervention using a speech-generating device in preschool children at risk for autism

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Abstract

Objective: To investigate whether preschool children at risk for autism would learn to request a preferred object using a speech-generating device (SGD) following a video modelling (VM) intervention and generalize requesting behaviours to preferred objects across stimuli and communication partners. **Method:** A single-subject multiple baseline design across participants ($n = 3$) was used to assess the effects of VM intervention on requesting behaviour. Intervention consisted of having each participant view a VM of two adults demonstrating the requesting of a preferred object using the SGD. Additionally, a least-to-most-prompting technique was employed to facilitate the production of a correct response. **Results:** All three participants demonstrated the ability to request preferred objects following the intervention and were able to generalize the newly acquired requesting behaviour across stimuli and people. **Conclusions:** Study findings indicate that a VM treatment package is effective in facilitating communication in children with little or no functional speech.

Keywords

Autism, communication skills, preschool, speech generating device, video modelling

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Introduction

Diagnostic parameters for autism spectrum disorders (ASD) include qualitative deficits in social interactions, communication, and/or restrictive interests with symptoms present before the age of three [1]. Although there is a great degree of variability in clinical and behavioural characteristics demonstrated by persons with autism, all persons with ASD have some type of communication impairment [2]. The language characteristics of ASD may include verbal perseveration, echolalia, poor prosodic features, idiosyncratic speech, poor non-verbal skills, a lack of spontaneous language initiation [3] and sharing [4], and deficits in pragmatic skills [5]. About half of the individuals with ASD will remain non-verbal or functionally non-verbal [4, 6, 7].

Many persons with autism with little or no functional speech rely on augmentative and alternative communication (AAC) methods to supplement or replace natural speech [8]. AAC aids such as speech-generating devices (SGDs) and text-to-speech software programs that convert words, digits, graphic symbols, and alphabets into speech have become increasingly available to persons with severe speech and language impairments [9]. Several studies have examined the

effects of SGDs as part of a treatment package on communicative and challenging behaviour in individuals with ASD [10–12]. Results indicate that the treatment packages that involve SGDs are effective in increasing the desired target behaviours in persons with ASD. Thus, the purpose of this study was to investigate whether preschool children with autism would learn to request a preferred object using an SGD following a video modelling (VM) intervention and further generalize requesting behaviour to preferred objects across stimuli and communication partners.

A number of visual approaches, such as imitative or modelling techniques, have been employed to enhance communication, behaviour, and social skills in children with ASD [13, 14]. One type of visual imitative technique is VM. VM intervention involves the video recording of a person who models appropriate behaviours followed by the presentation of the recorded video clip to the person with autism who would imitate the behaviour seen from VM [15–17]. According to Hine and Wolery [18], VM intervention consists of the following specific steps:

- (1) record appropriate behaviours to be imitated by the participant,
- (2) edit images of appropriate or new behaviour shown on a video monitor to the participants,
- (3) show clips of the appropriate target behaviour or several exemplars of the target behaviour to the participants multiple times,
- (4) practice the newly acquired skills in context, and

- (5) assess generalization of the new skills across settings, people, or stimuli.

There are several advantages in using VM as an intervention tool. First, VM capitalizes on the visual modality that has been observed to be a strength for children with autism [16, 18–20]. Second, extraneous stimuli can be removed and the effects of stimulus over-selectivity reduced by focusing on only the selected features of a targeted behaviour [21, 22]. Finally, VM interventions are based on the premise of observational learning that has been found to be an effective teaching strategy for persons with disabilities [23].

There are a number of studies that have investigated the effects of VM intervention on communication skills such as responding to questions, production of speech, and spontaneous requesting [21, 22, 24–29]. Specifically, Charlop-Christy et al. [21] studied the effects of VM on conversational skills in five students with ASD. Two scripted conversations were presented to participants for two independent conditions, *in vivo* (live, real-time models) and VM. Each conversation began with a question to which the participant was requested to provide a response followed by another question. The results indicated that in contrast to *in vivo* modelling, the VM conversation took less time to learn and the participants were able to generalize conversational skills across different activities, persons, and settings. Along similar lines, Charlop and Milstein [28] investigated whether participants would acquire, generalize, and maintain conversational skills following VM intervention. Results indicated that all three participants reached criterion and generalized newly acquired skills across conversational partners, settings, and conversation types (i.e. concrete and abstract). In addition, Sherer et al. [29] targeted conversational skills in five children with autism utilizing either ‘self’ or ‘other’ as the VM and found that three of the five children acquired the skills and generalized the scripted conversations or semantic equivalents across settings and partners.

Furthermore, Buggey et al. [27] investigated the effect of a video self-model (VSM) paradigm on the verbal behaviour of three participants with autism. VSM involved participants viewing their own taped behaviours. All three participants increased the frequency of their verbal responses during the interventions. Apple et al. [24] reported an increase in compliment-giving initiations and responses in children with autism using both adult and peer video models. In conclusion, previous research indicates that VM intervention has the potential to enhance the communication skills of individuals with ASD.

There are only a few studies that have investigated the effects of VM intervention on spontaneous requesting skills. Wert and Neisworth [22] examined spontaneous requesting skills in children with autism using VSM. Video recordings of four preschool participants with autism were obtained during a 30-min prompted play session in the home. The videos were then edited to show only unprompted requesting. The participants viewed the VSM before going to school. All four participants demonstrated an increase in spontaneous requesting in the school setting following the VSM. Three of the four participants demonstrated maintenance of spontaneous requesting. No generalization data were collected in the study, but were implied as training with the VSM occurred in one setting, at home, while data were collected in a second setting, at school.

The only other study that examined requesting skills using VM intervention was conducted by Banda et al. [30]. The investigators examined whether or not two individuals with autism would learn to use an SGD to request a preferred object after watching a VM. Participant one was a 17-year-old male with autism, severe speech and language impairment, and intellectual impairment. Participant two was a 21-year-old male with severe autism and severe speech and language impairment. The participants watched a 10–15 s VM demonstrating the operation of an SGD to request a preferred object. Both participants were able to demonstrate the use of the SGD to request a preferred object, but were unable to generalize requesting behaviour using the SGD to a second preferred object.

The present study, a systematic replication of the Banda et al. [30] study, investigated whether preschool children at risk for autism would learn to request preferred objects using an SGD following VM intervention. Specifically, this study examined the following four research questions: (a) what is the effect of VM intervention on the number of requests for preferred objects using an SGD? (b) What is the effect of VM intervention on the number of requests for untrained preferred objects using an SGD across stimuli and people? (c) What is the effect of VM intervention on the maintenance of requesting behaviours for preferred objects using an SGD? and (d) What is the effect of VM intervention on the perception of the effectiveness of treatment by the stakeholders?

Methods

Participants

The Texas Tech University Health Sciences Center Institutional Review Board approved the study and parental consent was obtained prior to conducting the study. The participants recruited for the study met the following criteria: age range of 18 months to 5 years, a diagnosis of autism or symptoms suggestive of autism, limited or no functional communication skills, an inability to appropriately request, and/or no spontaneous requesting ability, and an ability to watch TV or videos independently. Exclusion criteria included a diagnosis of hearing or visual impairment, an inability to demonstrate comprehension of the experimental stimuli, and a history of SGD use.

Preliminary assessments

The following tests were administered to all three participants who met the inclusion criteria: the Preschool Language Scales, fourth edition (PLS-4) [31], the Childhood Autism Rating Scale (CARS) [32] or the Modified Checklist for Autism in Toddlers (M-CHAT) [33], and the Communication and Symbolic Behavior Scales (CSBS) [34]. Overall developmental abilities were measured using the Differential Ability Scales (DAS) [35] or Developmental Profile 3 (DP-3) [36].

Participant profiles

Participant one. Participant one was a three-year, eight-month-old Caucasian male. He lived at home with both

parents and two older sisters. He was enrolled in an early childhood programme through the public school where he received developmental services, speech therapy, and home training. Prior to transitioning to the public school, he had been enrolled in a home-based early intervention programme. Participant one also received private speech therapy once a week. Participant one was initially classified as demonstrating Non-Categorical Early Childhood/Autism. This classification was later confirmed as autism. Preliminary evaluations rated his behaviours on the CARS as severely autistic. His performance on the DAS yielded a standard score of 44 (mean = 100, confidence interval = 44–59, percentile rank = <0.1). Participant one received standard scores of 50 on all areas of the PLS-4 indicating that his performance was below the normal range of 85–115.

Additional evaluation of communication skills using the CSBS [34] revealed that participant one's composite standard score was 7, which fell one standard deviation (SD) below the mean (mean = 10; SD = 3). Participant one's primary communication function was regulation. He used only two conventional gestures: giving and reaching. His primary communicative means were gestures with and without vocalizations. The parent also completed the Caregiver Perception Rating which indicated that participant one's behaviour during the administration of the CSBS was 'typical' or 'greater' than usually observed. Participant one had a recent history of some sign use (hand-over-hand 'more') but no SGD exposure or instruction. The parents also completed the interview portion of the CSBS. Parents reported that participant one communicated with sounds, gestures, and by handing objects to people. He did not produce any words but understood approximately 15 words. The parents reported that participant one would get frustrated when they could not figure out what he wanted.

Participant two. Participant two was a two-year, two-month-old Caucasian male. He lived with both parents and an older brother who had been diagnosed with Asperger Syndrome. Participant two was enrolled in an early childhood intervention programme which provided developmental and speech therapy services in the home. There was a history of sign use but no SGD use.

Preliminary evaluations rated his behaviours on the CARS as severely autistic. His performance on the DP-3 yielded a standard score of 60. This score was categorized as 'Delayed' (mean = 100, confidence interval = 52–68, percentile rank = 0.4). Participant two received the following standard scores on the PLS-4: Auditory Comprehension and Expressive Communication = 57; Total Communication = 52. These scores indicate delay in expressive and receptive communication.

Evaluation of communication skills using the CSBS indicated that participant two used a wider variety of conventional gestures: giving, pushing away, reaching, pointing, and head nodding. His primary means of communication was gestural with and without vocalization. The parent rated his behaviour during administration of the CSBS as 'typical' to 'less active' than usual. Parents also completed the interview portion of the CSBS and reported that participant two communicated by pointing, grunting, and gesturing. He did not produce any words but understood

approximately 24 words. He signed 'more' but only in imitation. The parents reported that participant two would 'throw a fit' when they could not figure out what he wanted.

Participant three. Participant three was a one-year, eight-month-old Hispanic male. He lived with his mother and an older brother and sister. Participant three was enrolled in an early childhood intervention programme which provided developmental and speech therapy services at the home. He enjoyed piling up his toys or lining them up on the coffee table. Sign use had been introduced and he was able to imitate signs, but did not produce them spontaneously. There was no history of access to an SGD.

Preliminary evaluations using the M-CHAT indicated failure on two critical items plus failure on seven additional items. His performance on the DP-3 yielded a standard score of 84 (mean = 100, confidence interval = 75–93, percentile rank = 14). Participant three received the following standard scores on the PLS-4: Auditory Comprehension = 67; Expressive Communication = 62; and Total Language Composite = 61. These scores indicate delay in expressive and receptive communication.

Participant three's primary communicative function on the CSBS was behaviour regulation. He also exhibited several communicative gestures such as giving, showing, pushing away, reaching, pointing, and waving. The parent commented that participant three's behaviour during administration was 'typical' to 'greater' than expected. The mother also completed the interview portion of the CSBS. The parent reported that participant three communicated by pointing, vocalizing, gesturing, and producing some single words. The mother reported that participant three produced four words (i.e. mama, wow, more, and tres, [three in Spanish]) and understood approximately 15 words. When the mother was unable to figure out what he wanted, participant three would cry, spin, or 'throw a fit'.

Setting

Baseline, intervention, generalization, and maintenance measures were obtained in the participant's homes during play or snack time and within the regular family routine. Parents chose the best placement for the SGD. For participant one, the SGD was placed on the bookshelf in front of the television; participant two on the fireplace; and, participant three on a small end table. Intervention occurred at a small dining table in the living room for participant one. Participant two and three received their intervention at the kitchen table. All sessions were video recorded for data analysis.

Materials

Preference probes

Likes and Dislikes Checklist [37] was used to determine a preliminary preference list (i.e. foods, drinks, and toys) for each participant. The investigator and parents then discussed each item on the list as a possible preferred/non-preferred stimulus item for intervention or generalization/maintenance. Potential preference items were presented and approach/non-occurrence behaviours measured [38]. The examiner

presented the preferred objects on the floor directly in front of the participant. If the participant pointed to, reached for, or grabbed the item within 5 s, the response was recorded as a preference and the participant was allowed brief access to it (30 s for edibles; 1–2 min for objects). If the participant demonstrated no response within 5 s, the lack of response was recorded as a non-preference. Criterion required an item to be chosen a minimum of 8 out of 10 presentations to qualify as 'preferred'.

Stimulus items

Preferred objects and non-preferred objects were digitally photographed for use on the Tech Talk, SGD (Advanced Multimedia Devices, Inc., Farmingdale, NY). Tech Talk is a low technology SGD that comes with multiple overlays. Each overlay can display eight picture symbols (i.e. 2×4 grid pattern) and an individualized message can be recorded and stored for each of those symbols. Digital photographs were chosen as stimuli because they are transparent and easily guessable [19]. Each participant was asked to identify the photograph of his/her preferred object in the following manner. The digital photograph of the preferred object was presented within an array of four photographs. The four photograph array consisted of the photograph of a preferred object and three photographs of non-preferred objects (i.e. any object or edible chosen at 20% or less during the preference assessment). All participants had to correctly identify the photographs of their preferred objects in 8 out of 10 trials. The order of preferred and non-preferred photographs was randomized across the 10 trials to avoid order effect.

The preferred stimulus items for participant one were M and Ms (90%), bubbles (90%), sand (90%), beans (80%), and rocks (80%). Non-preferred items included the light stick (20%), Tonka truck (20%), and Cheerios (20%). The preferred stimulus items for participant two included popcorn (90%), a fish ball (90%), bubbles (80%), a train (80%), and a grinder (80%). Non-preferred items were pickles (0%), broccoli (10%), and lemons (20%). The preferred stimulus items for participant three included a grinder (80%), bubbles (80%), M and Ms (80%), Tootsie Rolls (80%), and a pen light (80%). Non-preferred items were onions (0%), broccoli (20%), and celery (20%).

Video model

Three of the preferred objects for each participant were chosen randomly for intervention and separate vignettes were produced for each of the preferred objects. Each vignette consisted of two adult female models. The first adult activated the SGD to request the object. The second adult delivered the object requested along with verbal praise (i.e. 'Good job. Here is the X'). The target behaviour was repeated three times for each vignette. The camera angle for each vignette was adjusted in a way that eliminated all extraneous information by zooming in on only the salient features needed for the participant to complete the task. The recorded vignettes were burned onto a DVD in order to play them on a laptop computer. The duration of each vignette ranged from 19 to 38 s. The edited DVD models averaged 25.5 s in duration. This duration is within the maximum length of 30–40 s for

simple behaviours as recommended by Nikopoulos and Keenan [39].

Trainers

The first author served as the primary trainer for the study. A graduate student in speech-language pathology assisted with the determination of inter-observer reliability and procedural integrity.

Experimental design

The partially non-concurrent multiple baseline design across subjects was utilized to demonstrate experimental control and establish a functional relationship between the independent and dependent variables [40, 41]. Simultaneous baselines were obtained for participants one and two and a non-concurrent baseline was obtained for participant three. The two simultaneous baselines obtained for participants one and two provided control for the effects of history and other extraneous variables and the third non-concurrent baseline obtained for the participant three provided additional replication. The design included baseline, intervention, generalization, and maintenance phases.

Dependent variable

The dependent variable in this study was the production of a spontaneous requesting behaviour. A correct response was defined as the participant touching and/or pressing the photograph on the SGD, resulting in the activation of the speech output. The participant received a score of one for each correct response and a zero for an incorrect response.

Independent variable

The independent variable was the VM intervention in conjunction with a least-to-most prompt strategy.

Procedure

During baseline, the participants were observed during home play or snack time. A photograph of one of the participant's preferred objects was randomly placed on the SGD. No instruction, cueing or prompting was given. Data were collected on any activation of the device directed towards requesting the preferred object.

Once a stable baseline was achieved, intervention was scheduled with the first participant. Intervention started with the participant viewing the VM. After the VM was viewed, the SGD and the preferred object were placed in front of the participant. The preferred and non-preferred objects were randomly chosen for each session and the photographs representing objects were randomly placed on an SGD overlay. A 10-s pause was given to determine whether or not the participant would activate the SGD without prompting. If the participant selected the preferred object by touching and/or pressing the photograph on the SGD, resulting in speech output, the behaviour was considered a correct response. A correct response was also recorded if a participant corrected his initial incorrect response prior to the initiation of prompting by the investigator. If the participant grabbed the preferred object, touched and/or

pressed an unmarked frame on the device, a non-preferred object, or a photograph that did not result in speech output, least-to-most prompts were provided. Prompts were provided as follows: (1) the investigator stated the correct response (i.e. a verbal command), (2) if no response was made by the participant, the investigator demonstrated the response (i.e. model), (3) if no response was made by the participant, the investigator provided physical guidance (i.e. hand-over-hand prompting), and (4) prompts were faded when no longer needed for the occurrence of a correct response. After the participant activated the device, the participant was allowed brief access to the preferred object (30 s for edibles; 1–2 min for objects). This process was repeated until response criteria had been met (8 out of 10 correct responses per session, over 3 consecutive sessions) or 30 min had elapsed. Once the first participant reached criteria, the intervention began with the second participant.

Generalization data were collected in the same conditions as baseline except the participant was expected to request a non-trained preferred object with their parent(s) as their communication partner. Photographs of the preferred objects were randomly assigned to the overlay of the SGD. Generalization data were collected over three sessions at the home. The SGD was placed as follows: participant one, on the bookshelf in front of the television; participant two, on the fireplace; participant three, on a small end table. No prompting or cueing was provided. The device was simply displayed for the participant to make a request if desired. Each session consisted of a maximum of 10 requests or 30 min duration.

Maintenance data were collected over a 30-day time period post generalization. The SGD was available in the home and parents were requested to complete a weekly chart indicating the frequency of SGD use and daily availability. Parents were contacted by phone or email during maintenance in order to support their use of the SGD in the home.

Inter-observer agreement

Inter-observer agreement was determined by having a trained observer review the recorded video data and independently score the requesting behaviour. Agreement data were collected for each participant for at least 30% of the sessions across all phases of the design. Using a total agreement strategy, the number of agreements was divided by the number of agreements plus disagreements and then multiplied by 100 [42]. Inter-observer agreement was 98%.

Procedural integrity

Procedural integrity was also determined by having a trained observer review the recorded video data for each participant for at least one baseline, one intervention, and one generalization probe using a checklist of procedural steps that the investigator followed during baseline, intervention, and generalization sessions. Prior to reliability scoring, the investigator and trained observer reviewed the procedural integrity checklist and levels of prompting and viewed sample sessions for baseline, intervention, and generalization. The trained observer recorded 'yes' if the step was correctly followed by the investigator and 'no' if the step was

incorrectly followed. A unit-by-unit agreement ratio was used to calculate treatment integrity scores for each session [42]. Procedural integrity of 100% was obtained across three participants.

Social validity

A five question survey was distributed to all parents following the conclusion of the data collection. The questionnaire, based on a 5-point Likert Scale (1 = Strongly Disagree to 5 = Strongly Agree), was scored as a summative index [43]. The questionnaire included five questions that addressed: value of the research in the home, importance of requesting skills for children on the autism spectrum, use of a VM strategy as an intervention, use of an SGD with the children, and gains for the children from the study. The average response for each individual question was computed for data analysis and interpretation.

Results

Overall, the data presented demonstrated that all participants learned to use the SGD to request preferred objects following the VM intervention. Furthermore, generalization across preferred stimuli and people and maintenance of acquired behaviour was also observed. Results are presented in Figure 1.

Participant one

During baseline, no spontaneous, independent requests using the SGD were observed. Participant one primarily requested by handing, reaching for, or grabbing what he wanted. Other requesting behaviours for participant one included pulling-up his shirt to request 'tickle' and changing body position. He required 13 intervention sessions before criterion response was reached.

Participant two

During nine baseline probes, participant two did not demonstrate use of the SGD. He did, however, use other forms of requesting such as pointing and humming. He needed only four intervention sessions to reach criterion response.

Participant three

In three of the four baseline probes, participant three did not spontaneously make a request using the SGD. However, for the very first baseline probe, participant three touched the picture on the SGD and activated the speech output feature. Participant three primarily requested by pushing the button on a toy so that mum would play 'tickle' again. Intervention consisted of eight sessions. The performance of participant three was highly variable. Although he did not reach criterion response (i.e. 8 out of 10 correct responses per session, over 3 consecutive sessions), he did demonstrate acquisition of requesting behaviour (Figure 1). There was a clear contrast between baseline and intervention probes. Intervention was terminated following session eight and was followed by the collection of generalization data.

The second question of this study examined whether or not the acquired behaviour would generalize to untrained people

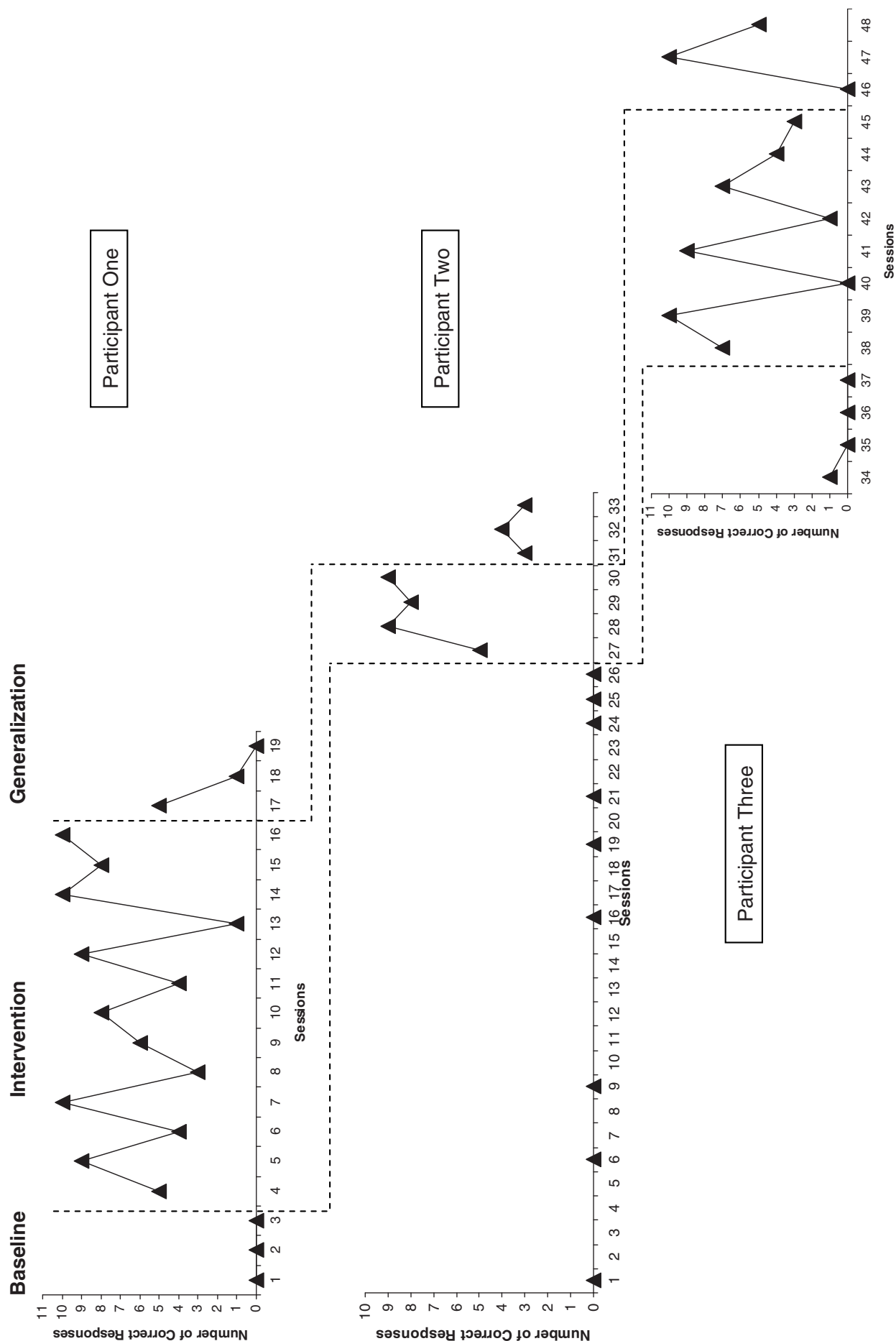


Figure 1. Percent correct responses across baseline, intervention, and generalization probes for three participants.

and untrained stimuli. The results were measured by adding the number of independent, non-cued responses over a total of three sessions. The parents were briefed on responding and not prompting or cueing a response during generalization. All three participants demonstrated generalization across people and stimulus items. Participant one used the SGD independently six times with his father as respondent. Participant two independently requested 10 times with his mother as respondent. Finally, participant three, although not reaching criterion during intervention, independently requested 15 times during generalization.

The third question investigated whether the participants would independently request preferred objects using the SGD in the 30-day period following generalization. Maintenance data were collected by the parent(s) who were asked to record the duration that the SGD was available per day, the number of days out of 30 days the device was available to them, and the number of uncued requests made by their child. During maintenance, participant one made a total of 115 requests over 50 h using the device for 28 of the 30 days. Participant two made a total of 84 requests over 12.5 h for 23 of the 30 days. Participant three made a total of 262 requests over 15 h using the device for all 30 days. In addition, participant three's mother reported that requesting with the SGD occurred in two different settings.

Percentage of non-overlapping data

Percentage of non-overlapping data (PND) were calculated by determining the number of non-overlapping data points between baseline and intervention phases for all three participants and were interpreted according to the criteria presented by Scruggs et al. [44]. These criteria describe PND greater than 90 as highly effective, PND between 70 and 90 as fairly effective, PND between 50 and 70 as questionable, and PND less than 50 as unreliable. Participant one obtained a PND score of 100%. The PND score for participant two was also 100%. Thus, the treatment was highly effective for participant one and two. Finally, the PND score for participant three was 75% indicating that the treatment was fairly effective in his case.

Social validity

Results on a five question survey indicated that the respondents felt the research was valuable (average = 4.3; range 4–5). Respondents also felt that appropriate requesting was important (5.0). Families felt VM was a somewhat useful intervention technique (average = 3.66; range 3–5), but the overall use of an SGD was strongly supported (average = 4.66; range 4–5). Finally, all respondents felt that the research had been beneficial for their children (average = 4.66; range 4–5).

Discussion

The purpose of this study was to investigate whether preschool children with autism would request preferred objects using an SGD following VM intervention. Data indicated that none of the participants were able to request objects using an SGD prior to intervention. All baseline measures were stable and indicated the absence of requesting

behaviour. The effectiveness of VM intervention was demonstrated by the acquisition of requesting behaviour in all three participants. PND results were particularly robust for participants one and two and acceptable for participant three.

Several other published single-subject design studies have examined the effect of VM intervention on the acquisition of various communication skills, including requesting in persons with autism [21, 22, 24, 26, 28–30]. Results indicate that successful outcomes using the VM intervention and are consistent with the findings of the current study. Banda et al. [30] and Wert and Neisworth [22] were the only studies that targeted spontaneous requesting as the dependent variable. However, the current study differs in several ways from both that of Banda et al. and Wert and Neisworth. First, in the Wert and Neisworth study, the participants demonstrated appropriate requesting skills prior to intervention; however, the requesting behaviour was not frequently observed. Participants in the current study and the Banda et al. study only requested inappropriately (i.e. pulling adults to the desired objects, grabbing/taking the desired objects) or ineffectively (i.e. whining, grunting, screaming, or having tantrums) until caregivers figured out what they wanted and supplied the desired object or activity. Banda et al. trained two individuals with severe disabilities who may have had years of reinforcement of non-acceptable requesting behaviours, thus making appropriate requesting less desirable for them than techniques already proven to elicit a desired response, albeit, unconventional. The participants in the current study had also experienced successful requesting using inappropriate behaviours; however, because of their young age, the inappropriate behaviours were relatively easy to modify and replace with appropriate requesting behaviours. Another difference was that the Banda et al. study established relatively less control over the stimulus items. Many intervention items remained in the classroom and therefore may have been offered without requiring a contingent response. Participants may have been cognizant of waiting for non-contingent opportunities to retrieve their preferred objects/activities. In the current study, all items remained under the investigators control. Additionally, in contrast to the current study, no prompting strategy was used by Wert and Neisworth [22] or Banda et al. [30]. The current study used a least-to-most prompt strategy as part of the treatment package.

The VM intervention in the current study was highly effective for at least two of the three participants (i.e. PND of 100%). Several of the following factors may have contributed to the results. First, clarity of the model in the videos may have contributed to the increase in requesting behaviour [21, 22]. The video clips shown to the participants limited the information only to the salient features needed to perform the task. That is, the camera narrowed in on the SGD, the hand of the model pressing the picture and activating the voice output feature, and the other hand delivering the requested object or activity. Second, the target behaviour was repeated three times on each vignette. Only one example of the target behaviour was presented on the VM in the Banda et al. study [30]. Third, VM is a visually advantageous learning mode for persons with autism. There is strong evidence that

persons with autism respond favourably to visual modality instruction [16, 18–20].

The effectiveness of treatment in this study may also have been due to the use of highly preferred items. Pace et al. [38] recommended preference assessments to increase the likelihood that participants would request items multiple times without satiation. In the current study, most items retained their preferred status over time. Participants were sometimes easily distracted by other stimuli but did maintain a desire to continue requesting those items which had been determined to be highly preferential. Conversely, those items which were not deemed as preferential were rarely, if ever chosen. In addition, the current study controlled the preferred items by having the investigator maintain possession of them. This limited the families providing items on a non-contingent basis outside of the study.

A related factor that may have played a role in increasing the requesting behaviour was the randomization of the preferred items and the inclusion of prompting in the treatment package. Three preferred items were randomly presented during intervention. Two others were held back for the generalization phase. Items were also randomly presented during the maintenance phase. The purpose of randomizing was to prevent or at least delay satiation. Deprivation and satiation work in concert with one another. Withholding an item increases the participant's desire (deprivation). Once the desire is satisfied, satiation occurs and the stimulus item is no longer reinforcing. The result can be a reduction in the person's cooperation and attention [45].

The current study, like several others [21, 23, 28, 46], utilized prompts to increase the likelihood of correct responding. Since prompting resulted in errorless responses, unconventional communication was extinguished and replaced by requests using an SGD. Supplementary instruction such as prompting, cueing, and gradual fading may be essential to teaching requesting skills [47] and must be considered in an AAC intervention.

All three participants, to varying degrees, were able to generalize requesting to untrained preferred objects and to the parents as respondents. According to Mirenda [2], intervention should only be considered successful if skills result in spontaneous, functional communication across settings and people. Thus, the results of this study indicate that requesting behaviour acquired as a result of VM intervention generalized across stimuli and conversational partners. Furthermore, all participants maintained independent, spontaneous use of the SGD during the maintenance period. The most positive aspect of maintenance was that it occurred in the participants' natural environment and with the primary caregivers as respondents.

Results of a social validity measure indicated that the parents strongly felt that their children had benefitted from the intervention. In addition, they agreed that their children should be able to indicate their wants and needs using an appropriate communicative behaviour. During the interview portion on the CSBS, all of the parents had reported frustration over their inability to 'guess' what their children were communicating. In addition, parents also reported that inappropriate behaviours such as crying and tantrums resulted when communication broke down. They all supported the use

of an SGD because it improved their comprehension of the child's intended message.

In summary, the general findings from this study suggest that VM can be an effective tool in the intervention programme for persons with autism. However, further research needs to be conducted to test the efficacy of VM with a large cohort of participants and across a variety of communicative behaviours.

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Declaration of interest

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of this paper.

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